COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	7/12/26	**DATE SUBMITTED:**	7/12/26
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ACTIVITY NO. _1__ LABORATORY ACTIVITY

1. PROCEDURE OUTPUT

5. Procedure

ILO A: Create Code That Follows the Basic C++ Code Structure

For this activity, you have to demonstrate the use of a **function prototype**. The section on class declaration and method definition will be used for the function prototype and the function will be defined in the follow method definition section after the main function.

A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

- Create a function that will take two numbers and display the sum.
- Create a function that will return whether variable A is greater than variable B.
- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Note:
- The driver program must call each function.
- The definitions must be after the main function.

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

In this section, the initial implementation for a class **triangle** will be implemented. The step-by-step procedure is shown below:

Step 1. Include the necessary header files. For this one, we only need `#include <iostream>`

Step 2. Create the triangle class. Assign it with private variables: totalAngle, angleA, angleB, and angleC.

```
class Triangle{
private:
    double totalAngle, angleA, angleB, angleC;
```

Step 3. We then create public methods. The constructor must allow for creation of the object with 3 initial angles to be stored in our previously defined variables `angleA`, `angleB` and `angleC`. Another method has to be made if the user wants to change the initial values, this will also accept 3 arguments to change the values in `angleA`, `angleB` and `angleC`. Lastly, a function to validate whether the given values make our shape an actual triangle.

```
public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};
```

Step 4. Define the methods.

```
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
```

```
        angleB = B;
        angleC = C;
        totalAngle = A+B+C;
    }

    void Triangle::setAngles(double A, double B, double C) {
        angleA = A;
        angleB = B;
        angleC = C;
        totalAngle = A+B+C;
    }

    const bool Triangle::validateTriangle() {
        return (totalAngle <= 180);
    }
```

Step 5. Create the driver code.

```
int main(){
    //driver code
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
    }

    return 0;
}
```

Include the output of running this code in section 6. Note your observations and comments.

6. Output

Sections	Answer
Header File Declaration Section	
Global Declaration Section	
Class Declaration and Method Definition Section	
Main Function	
Method Definition	

Table 1-1. C++ Structure Code for Answer

Table 1-2. ILO B output observations and comments.

ILO C: Solve Different Problems using the C++ Programming Language

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.
2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
3. Create a C++ program that has a function that will calculate the distance between two points.
4. Modify the code given in ILO B and add the following functions:
    a. A function to compute for the area of a triangle
    b. A function to compute for the perimeter of a triangle
    c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

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ILO - A

Programiz C++ Online Compiler

main.cpp

Share Run

```cpp
#include <iostream>
using namespace std;

// Function Prototype
void displaySum(int a, int b);

int main()
{
    int num1, num2;

    cout << "Enter two numbers: ";
    cin >> num1 >> num2;

    displaySum(num1, num2);

    return 0;
}

// Function Definition
void displaySum(int a, int b)
{
    cout << "Sum = " << a + b << endl;
}
```

Output

```
Enter two numbers: 13
12
Sum = 25


=== Code Execution Successful ===
```

main.cpp

Share Run

```cpp
#include <iostream>
using namespace std;

// Function Prototype
bool isAGreaterThanB(int A, int B);

int main()
{
    int A, B;

    cout << "Enter value for A: ";
    cin >> A;

    cout << "Enter value for B: ";
    cin >> B;

    if (isAGreaterThanB(A, B))
        cout << "A is greater than B." << endl;
    else
        cout << "A is not greater than B." << endl;

    return 0;
}

// Function Definition
bool isAGreaterThanB(int A, int B)
{
    return A > B;
}
```

Output

```
Enter value for A: 55
Enter value for B: 36
A is greater than B.


=== Code Execution Successful
```

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main.cpp

```cpp
#include <iostream>
using namespace std;

// Function Prototype
bool logicalOperations(bool x, bool y);

int main()
{
    bool value1, value2;

    cout << "Enter first Boolean value (0 or 1): ";
    cin >> value1;

    cout << "Enter second Boolean value (0 or 1): ";
    cin >> value2;

    if (logicalOperations(value1, value2))
        cout << "Logical operations completed successfully." << endl;

    return 0;
}

// Function Definition
bool logicalOperations(bool x, bool y)
{
    cout << "\nLogical Operations:" << endl;
    cout << "x AND y = " << (x && y) << endl;
    cout << "x OR y  = " << (x || y) << endl;
    cout << "NOT x   = " << (!x) << endl;
    cout << "NOT y   = " << (!y) << endl;
    cout << "x XOR y = " << (x != y) << endl;

    return true;
}
```

Output

```
Enter first Boolean value (0 or 1): 0
Enter second Boolean value (0 or 1): 1

Logical Operations:
x AND y = 0
x OR y  = 1
NOT x   = 1
NOT y   = 0
x XOR y = 1
Logical operations completed successfully.


=== Code Execution Successful ===
```

ILO - B

Programiz C++ Online Compiler

main.cpp

```cpp
#include <iostream>
using namespace std;

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    // Constructor
    Triangle(double A, double B, double C);

    // Setter
    void setAngles(double A, double B, double C);

    // Validator
    bool validateTriangle();
};

// Constructor Definition
Triangle::Triangle(double A, double B, double C)
{
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

// Setter Definition
void Triangle::setAngles(double A, double B, double C)
{
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

Output

```
The shape is a valid triangle.


=== Code Execution Successful ===
```

main.cpp

```cpp
    angleC = C;
    totalAngle = A + B + C;
}

// Setter Definition
void Triangle::setAngles(double A, double B, double C)
{
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

// Validation Definition
bool Triangle::validateTriangle()
{
    return (angleA > 0 && angleB > 0 && angleC > 0 && totalAngle == 180);
}

// Driver Program
int main()
{
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle())
    {
        cout << "The shape is a valid triangle." << endl;
    }
    else
    {
        cout << "The shape is NOT a valid triangle." << endl;
    }

    return 0;
}
```

Output

```
The shape is a valid triangle.

=== Code Execution Successful ===
```

ILO - C

Programiz C++ Online Compiler

main.cpp

```cpp
#include <iostream>
using namespace std;

// Function Prototype
void swapNumbers(int &a, int &b);

int main()
{
    int num1, num2;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    cout << "\nBefore swapping:" << endl;
    cout << "First Number = " << num1 << endl;
    cout << "Second Number = " << num2 << endl;

    swapNumbers(num1, num2);

    cout << "\nAfter swapping:" << endl;
    cout << "First Number = " << num1 << endl;
    cout << "Second Number = " << num2 << endl;

    return 0;
}

void swapNumbers(int &a, int &b)
{
    int temp = a;
    a = b;
    b = temp;
}
```

Output

```
Enter first number: 100
Enter second number: 500

Before swapping:
First Number = 100
Second Number = 500

After swapping:
First Number = 500
Second Number = 100


=== Code Execution Successful ===
```

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main.cpp

Share Run

```cpp
#include <iostream>
using namespace std;

// Function Prototype
double kelvinToFahrenheit(double kelvin);

int main()
{
    double kelvin;

    cout << "Enter temperature in Kelvin: ";
    cin >> kelvin;

    cout << "Temperature in Fahrenheit: "
         << kelvinToFahrenheit(kelvin) << endl;

    return 0;
}

double kelvinToFahrenheit(double kelvin)
{
    return (kelvin - 273.15) * 9 / 5 + 32;
}
```

Output

```
Enter temperature in Kelvin: 32
Temperature in Fahrenheit: -402.07


=== Code Execution Successful ===
```

Programiz C++ Online Compiler

main.cpp

Share Run

```cpp
#include <iostream>
#include <cmath>
using namespace std;

// Function Prototype
double distance(double x1, double y1, double x2, double y2);

int main()
{
    double x1, y1, x2, y2;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    cout << "Distance = "
         << distance(x1, y1, x2, y2) << endl;

    return 0;
}

double distance(double x1, double y1, double x2, double y2)
{
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
}
```

Output

```
Enter x1: 54
Enter y1: 63
Enter x2: 45
Enter y2: 10
Distance = 53.7587


=== Code Execution Successful ===
```

main.cpp

Share Run

Output

```
#include <iostream>
#include <cmath>
using namespace std;

class Triangle
{
private:
    double sideA, sideB, sideC;

public:
    Triangle(double A, double B, double C);

    void setSides(double A, double B, double C);

    bool validateTriangle();

    double area();

    double perimeter();

    void triangleType();
};

// Constructor
Triangle::Triangle(double A, double B, double C)
{
    sideA = A;
    sideB = B;
    sideC = C;
}

// Setter
void Triangle::setSides(double A, double B, double C)
{
    sideA = A;
```

```
Valid Triangle
Perimeter = 12
Area = 6
Triangle Type: Others (Right Triangle)


=== Code Execution Successful ===
```

Programiz C++ Online Compiler

main.cpp

Share Run

Output

```
    sideB = B;
    sideC = C;
}

// Validate Triangle
bool Triangle::validateTriangle()
{
    return (sideA + sideB > sideC &&
            sideA + sideC > sideB &&
            sideB + sideC > sideA);
}

// Perimeter
double Triangle::perimeter()
{
    return sideA + sideB + sideC;
}

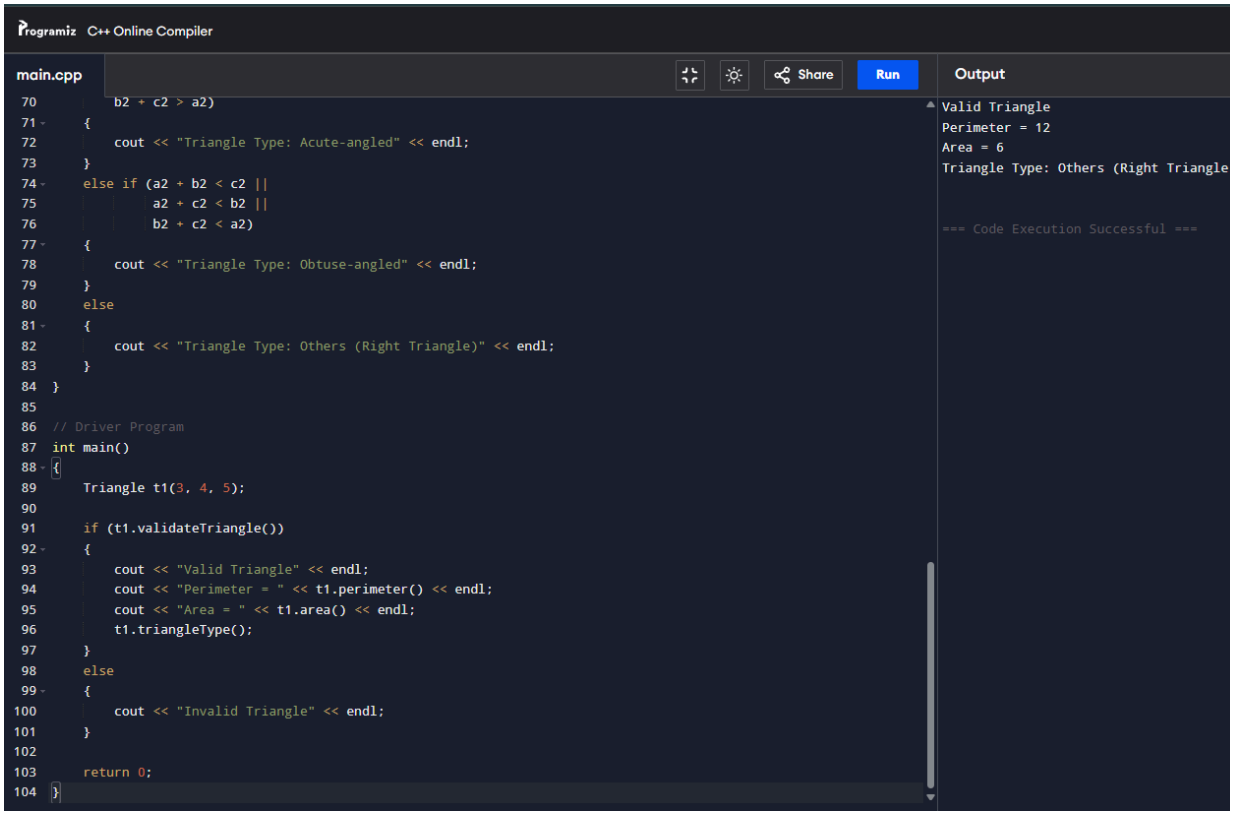
// Area (Heron's Formula)
double Triangle::area()
{
    double s = perimeter() / 2;
    return sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

// Triangle Type
void Triangle::triangleType()
{
    double a2 = sideA * sideA;
    double b2 = sideB * sideB;
    double c2 = sideC * sideC;

    if (a2 + b2 > c2 &&
        a2 + c2 > b2 &&
        b2 + c2 > a2)
```

```
Valid Triangle
Perimeter = 12
Area = 6
Triangle Type: Others (Right Triangle)


=== Code Execution Successful ===
```



SECTIONS	
Header File Declaration Section	#include <iostream>
Global Declaration Section	using namespace std;
Class Declaration and Method Definition Section	Declares the triangle class, its private data members (totalAngle, angleA, angleB, angleC), and the public constructor and member function prototypes.
Main Function	Creates a Triangle object, calls the validateTriangle() function, and displays whether the triangle is valid.
Method Definition	Defines the constructor, setAngles(), and validateTriangle() functions that initialize, modify, and validate the triangle's angles.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **Chatgpt,grammarly,microsoft copilot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar and language checking,code assistance and debugging,clarification of C++ programming concepts,formatting and organization of the document**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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